

Foundation of Programming

Assignment No. 14

PROBLEM STATEMENT:

Write a C program to input a binary number from the user and convert it into a decimal number.

OBJECTIVE:

1. To understand number system conversions.
2. To learn how to convert binary numbers to decimal numbers in C.
3. To practice the use of loops and arithmetic operations.
4. To develop logical thinking for number system problems.

THEORY:

A binary number system uses only two digits: 0 and 1. Each digit of a binary number represents a power of 2 depending on its position. To convert a binary number into its decimal equivalent, each digit is multiplied by 2 raised to the power of its position and the results are added together. In C programming, this conversion can be implemented using loops and arithmetic operations.

ALGORITHM:

```
STEP 1:    START
STEP 2:    Declare variables n, b, d = 0, c = 0
STEP 3:    Accept the value of b from the user
STEP 4:    Run a loop till b != 0
            Store the remainder of b divided by 10 in d
            If d is neither 1 nor 0, display invalid input
            Else
                Store the sum of n and d * 2 raised to c
                Increment c
                b = b / 10
STEP 5:    Display n
STEP 6:    STOP
```

PLATFORM : Linux

CODE:

```
1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      long long binary;
6      int decimal = 0, i = 0, remainder;
7
8      printf("Enter a binary number: ");
9      scanf("%lld", &binary);
10
11     while(binary != 0) {
12         remainder = binary % 10;        // get last digit (0 or 1)
13         decimal += remainder * pow(2, i);
14         binary = binary / 10;           // remove last digit
15         i++;
16     }
17
18     printf("Decimal equivalent = %d\n", decimal);
19
20     return 0;
21 }
```

OUTPUT:

```
Enter a binary number: 11011
Decimal equivalent = 27
```

CONCLUSION:

Thus, the C program successfully converts a binary number entered by the user into its decimal equivalent using loops and arithmetic operations.